

Intro

It seems that overnight Generative Artificial Intelligence (GenAI) have made their way into schools and learning institutions. As administrators scramble to implement and enforce policies it is important to critically analyze the impact on students and the possible benefit of AI on students. The dataset analyzed 50,000 students, collecting data on grade point average (GPA), self-reported ratings of anxiety and knowledge as well as behavioral patterns.

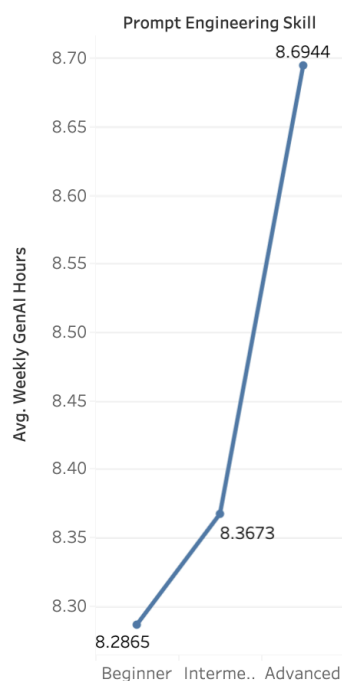
Objective

The purpose of this analysis project was to investigate and analyze how AI usage affects student success and find any correlation between success and existing policies. The key metrics that were used were the difference in pre and post semester GPAs, self-reported student anxiety and skill retention.

Tools Used

Excel, Tableau

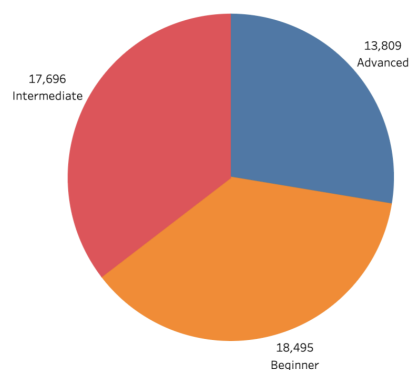
Visualizations and Insights



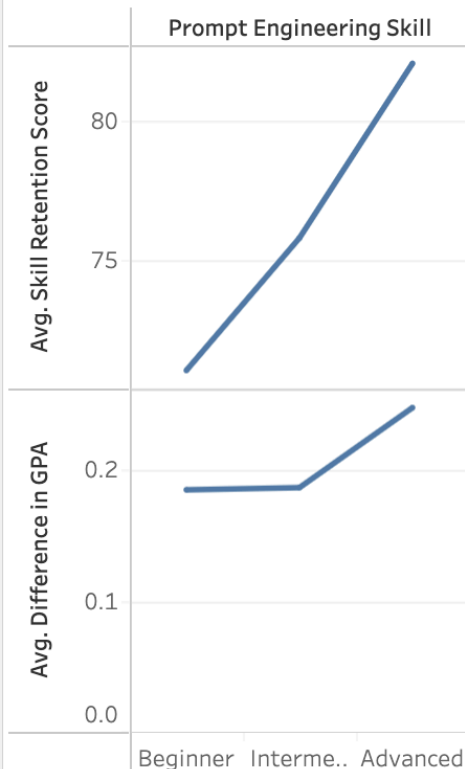
1.1 The classification of skill level (beginner, intermediate, and advanced) was a self-assessed proficiency in crafting effective prompts. In regards to this measure students that self identified as advanced had an average weekly usage of 8.7 hours, intermediate with 8.4, and beginner 8.3.

1.2 The breakdown of the students surveyed is below with most respondents self-identifying as being at an intermediate level of prompt creation.

Share of Students Across Skill Levels

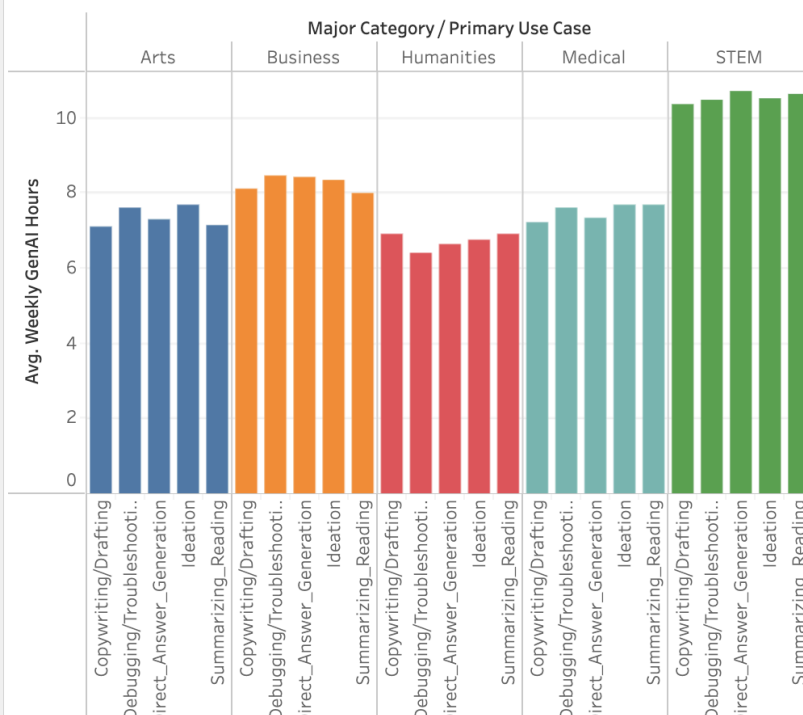


Prompt Engineering Skill Compared to Skill Retention and Changes in GPA

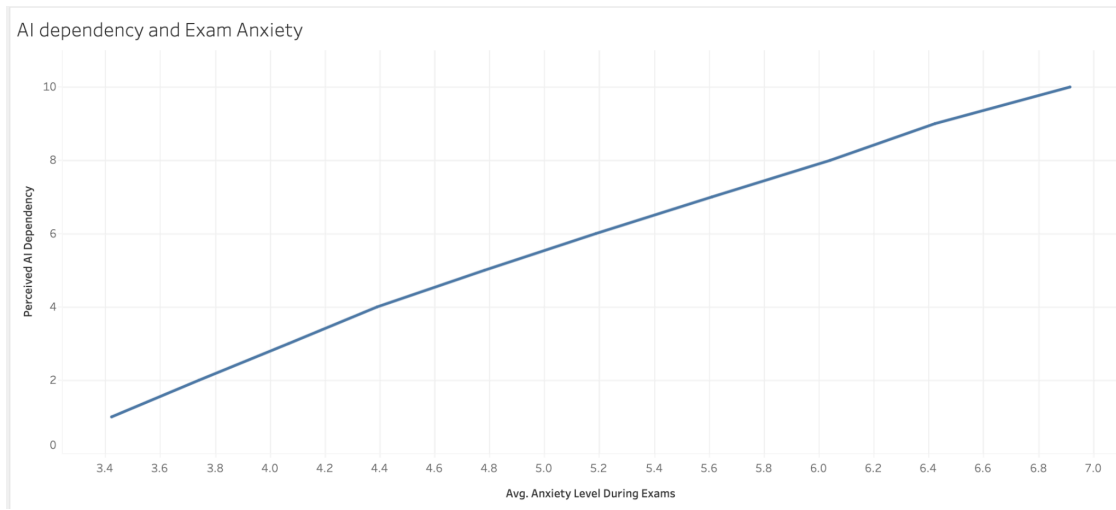


1.3 The two graphics display that the students that on average scored the highest on the post-semester test that measured skill retention and the students with the largest increase in grade point average (GPA) on average were students that self described as being at an advanced level of prompt generating. As we saw from the above visualizations, these students also on average used AI tools the most.

Generative AI Usage by Major and Use Case



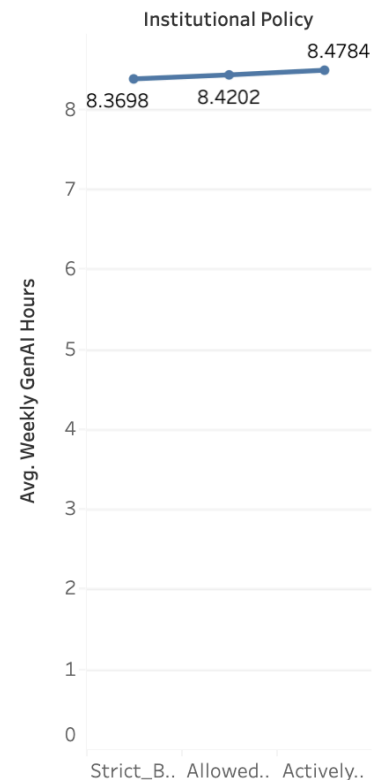
1.4 In the corresponding graphic we can see that STEM majors use Generative AI tools the most (spend the most hours weekly)



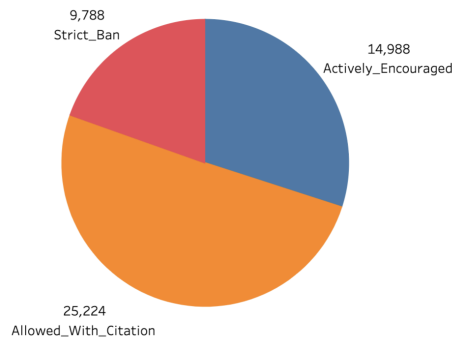
1.5 The above graphic shows that students with perceived AI dependency tend to have higher levels of anxiety during exams. Anxiety level, self-reported exam anxiety score from 1 (minimal) to 10 (severe), increasing could suggest that students' reliance on AI may not be beneficial in the long run as it may help students complete assignments faster. It is detrimental to students' ability to learn and digest information for later recall.

1.6 Institutions attended by the surveyed students were also recorded into three categories in regards to the institution's official AI policy: strict ban, allowed with citation and actively encouraged. What we see in the graph to the right is that the schools policy does not have a considerable impact on the actual usage of generative AI.

Actual Usage vs. Policy



Institutional Policy



Conclusions and Recommendations

While usage might not necessarily translate to increased success, students with more advanced prompt engineering skills had the highest increase in gpa over the course of the measured semester. Conversely, among students surveyed those that had the highest usage of AI tools had the most anxiety when it came to exams and testing. We can most likely assume that this anxiety could be in part due to the fact that exams test students' knowledge, recall and ability to apply learned concepts. It may be easy to draw the conclusion that students that use AI tools more had higher skill retention scores-meaning that AI use had a positive impact on scores. However, it is important to consider that a possible explanation for this correlation could be that students that possess strong prompt engineering skills also possess strong skill retention skills.

Moving forward I would recommend that institutions look to implement tools that detect AI, specifically institutions that seek to implement strict AI policies. Furthermore, assignments that challenge students' application of concepts would better evaluate students' knowledge of the course material .